

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Subject:	Cryptology (PEC-CSBS801A)	Semester:	8th Semester
Year:	2025-2026	Exam:	End Semester Examination
UPID:	008440	Paper Code:	CS/B.TECH(N)/EVEN/ SEM-8/8440/2025-2026/1028

Time Allotted: 3 Hours

Full Marks: 70

The figures in the margin indicate full marks. Candidates are required to give their answers in their own words as far as practicable.

GROUP-A (VERY SHORT ANSWER TYPE QUESTION)

Answer any ten of the following:

[1 × 10 = 10]

- (I) What is a substitution technique in classical encryption? **[1]**
- (II) What is modular arithmetic? **[1]**
- (III) Name one asymmetric key exchange protocol. **[1]**
- (IV) Name one digital signature standard. **[1]**
- (V) What is PGP? **[1]**
- (VI) What is the need for security at multiple levels? **[1]**
- (VII) Differentiate differential and linear cryptanalysis. **[1]**
- (VIII) What is primality testing? **[1]**
- (IX) Differentiate authentication function and MAC. **[1]**
- (X) Differentiate intruders and malicious software. **[1]**
- (XI) Differentiate legal, ethical, and professional aspects of security. **[1]**
- (XII) Explain the strength of DES. **[1]**

GROUP-B (SHORT ANSWER TYPE QUESTION)

Answer any three of the following:

[5 × 3 = 15]

2. Briefly explain primes and primality testing. [5]
3. Briefly explain authentication protocols. [5]
4. Explain intruders and malicious software. [5]
5. Describe the OSI security architecture with one example. [5]
6. Evaluate the effectiveness of firewalls against modern threats. [5]

GROUP-C (LONG ANSWER TYPE QUESTION)

Answer any three of the following:

[15 × 3 = 45]

7. (a) Describe symmetric key ciphers, S-DES, DES principles, and strength of DES. [8]
(b) Outline block cipher modes of operation. [7]
8. (a) Describe the principles of the following asymmetric key ciphers: RSA and ElGamal. [8]
(b) Outline different key distribution and management techniques. [7]
9. Describe different authentication protocols and digital signature protocols with suitable examples. [15]
10. Discuss how firewalls and antivirus counter malicious software and intruders. [15]
11. Critically compare OSI security architecture with the network security model. [15]